PAVI: Bootstrapping Accountability and Privacy to IPv6 Internet

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***Abstract*— Accountability and privacy are considered valuable but conflicting properties in the Internet, which at present does not provide native support for either. Past efforts to balance accountability and privacy in the Internet have unsatisfactory deployability due to the introduction of new communication identifiers, and because of large-scale modifications to fully deployed infrastructures and protocols. The IPv6 is being deployed around the world and this trend will accelerate. In this paper, we propose a private and accountable proposal based on IPv6 called PAVI that seeks to bootstrap accountability and privacy to the IPv6 Internet without introducing new communication identifiers and large-scale modifications to the deployed base. A dedicated quantitative analysis shows that the proposed PAVI achieves satisfactory levels of accountability and privacy. The results of the evaluation of a PAVI prototype show that it incurs little performance overhead, and is widely deployable.**

***Index Terms*— Accountability, privacy, IPv6.**

I. Introduction

ACCOUNTABILITY and privacy are both considered valuable in the Internet, although they appear to conflict. However, the Internet does not at present provide native support for either. On the one hand, the lack of accountability makes it possible for attackers to spoof source addresses to launch massive distributed denial of service attacks. Network operators cannot stop ongoing attacks because they do not know the true senders and thus cannot effectively filter spoofing traffic. On the other hand, people have the right to anonymously access information through the Internet. Unfortunately, the disclosure of pervasive monitoring and mass surveillance [2]–[5] causes people to become increasingly concerned about their privacy, and to seek applications such as onion routing [6], [7] and mix network [8] to protect it. However, these techniques are usually heavyweight and often degrade user experience.

A considerable amount of research has been conducted on accountability and privacy in the last few decades. However, the mechanisms proposed in this context often ensure that one of the two properties is obtained at the cost of the other. The Accountable and Private Internet Protocol (APIP) [9], and the Accountable and Private Network Architecture (APNA) [10] are the main protocols that attempt to strike a balance between accountability and privacy at the network layer. However, both protocols have limitations (see more details in Section VIII), the most serious one of which is that both introduce new identifiers to create communication connections that require large-scale modifications to existing Internet infrastructures and protocols.

Under this circumstance, we ask the question: *"Can we bootstrap accountability and privacy to the current Internet without introducing new communication identifiers and large-scale modifications to fully deployed infrastructures and protocols?"* We are interested in this kind of approach because it can deliver practical benefits without having to build everything from scratch. More specifically, we have reasons to believe that because of the following facts:

- IP addresses as the current communication identifiers are closely related to the Internet infrastructure, including DNS, routing protocols. Once the Internet adopts new communication identifiers to replace IP addresses, all protocols, infrastructures, mechanisms, or applications that rely on IP addresses need to be updated or modified.
- IPv6 has sufficient address space to allow for user designs. Currently, the IPv6 is being deployed increasingly commonly around the world, and this trend will accelerate. Approximately 30% of users access Google over IPv6 [11]. Belgium regularly delivers more than 50% of traffic to major content providers over IPv6 [12]. The Internet Engineering Task Force (IETF) will stop requiring IPv4 compatibility in new or extended protocols, and its subsequent work will optimize for and depend on IPv6 [13].

In this paper, we propose a new private and accountable proposal based on IPv6 called PAVI[1] that bootstraps accountability and privacy to the IPv6 Internet. We consider there are two key challenges in bootstrapping accountability and privacy to the existing Internet. The first one is how to associate a sender's identity with its packets in a lightweight and privacy-preserving manner, and the second one is how to do so without large-scale modifications to the existing Internet.

PAVI uses three novel designs to address these challenges. First, it uses IPv6 addresses instead of creating new communication identifiers, unlike APIP and APNA. Currently, IPv6 interface identifiers are allowed to be generated flexibly, e.g.,

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[1]The Arabic number 6 corresponds to the Roman numeral VI.

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EUI-64 [14], temporary address [15], CGA [16], semantically opaque interface identifiers [17]. Therefore, PAVI defines a new IPv6 address generation mechanism, accountable privacy address (APA), which associates user identities with IPv6 addresses by encrypting user identifiers to generate IPv6 interface identifiers. This makes each data packet contain a sender identity as previous arts do without introducing new communication identifiers and sacrificing sender privacy.

Second, PAVI takes full advantage of IPv6 addresses to protect privacy by assigning multiple APAs to each host. As IPv6 address space is large enough, the current usage of IPv6 addresses is not sufficient. A host with multiple APAs can determine using which address to communicate with the peer. Moreover, APAs are usually short-lived, which makes it difficult for attackers to undermine user privacy.

Third, PAVI uses standardized and well-maintained protocols with well-designed extensions as much as possible to simultaneously achieve accountability and privacy. This feature allows PAVI to avoid large-scale modifications to infrastructures and protocols related to communication identifiers, and render it widely deployable, unlike APIP and APNA. Also, the forwarding device in PAVI only uses symmetric cryptographic operations without degrading its performance, which is more efficient than using asymmetric cryptographic operations to verify public key infrastructure (PKI) certificates when APIP adapts to today's Internet.[2]

Contributions. We make the following contributions in this paper:

- We propose PAVI that bootstraps accountability and privacy to the IPv6 Internet without introducing new communication identifiers and large-scale modifications to fully deployed infrastructures and protocols.
- We conduct a quantitative analysis to show that PAVI satisfies the demands of accountability.
- We validate the privacy goals of PAVI using information-theoretic entropy.
- We demonstrate the practicality and deployability of PAVI through the performance evaluation of a prototype. The results show that PAVI is a lightweight and deployable protocol.

The remainder of this paper is organized as follows: Section II presents the problem formulation, and Section III presents an overview of PAVI. Section IV describes the details of PAVI. We detail a protocol analysis of PAVI in Section V, and describe its implementation and performance evaluation in Section VI. Some deployment issues are discussed in Section VII. Related work is reviewed in Section VIII. Finally, we offer the conclusions of the paper in Section IX.

II. Problem Formulation

Our goal is to bootstrap accountability and privacy in the IPv6 Internet in a lightweight and deployable manner. In this section, we describe the motivation, the resources and protocols available, the design goals, and the adversary model.

[2]In Section 7.3 of APIP [9], PKI certificates are used to verify delegates' identities before a verifier trusts their response. Communication with delegates and verification of PKI certificates will reduce the performance of the verifier.

A. Motivation

Currently, APIP [9] and APNA [10] are state of the arts in balancing accountability and privacy in the network. However, both require large-scale modifications to today's Internet infrastructures and protocols, which renders them difficult to be widely deployable. As both APIP and APNA introduce new communication identifiers, protocols and infrastructures based on IP addresses need to be updated, e.g., routing protocols and DNS. Although they provide methods to be deployed on today's Internet, there are still some limitations. More specifically, when APIP adapts to today's Internet, the introduction of PKI certificates and the extra communication with delegates at routers (verifiers) will reduce verifiers' performance (see more details in Section VI-B.4). For APNA, unless an AS fully adopts APNA, it cannot mitigate the privacy leakage of host addresses observed by an adversary. In addition, even if all routers in the AS forward packets based on EphID, each router should maintain routing information to the EphIDs in the AS. For those large ISPs, each router may need to handle millions of flat entries, which would become burdensome for routers. Therefore, we aim to design a protocol that can bootstrap accountability and privacy to today's Internet without introducing new communication identifiers and large-scale modifications to existing infrastructures and protocols.

B. What We Have

The large IPv6 address space, address usage practices, and existing standardized and well-maintained protocols can be used to bootstrap accountability and privacy to today's IPv6 Internet.

1) Large IPv6 Address Space: The address space of IPv6 is considerably larger than that of IPv4. For even a single /64 IPv6 prefix, the address space is 2^{32} times as large as the entire IPv4 address space. When using stateless address autoconfiguration (SLAAC) [18], a host generally configures two global unicast addresses within a /64 IPv6 prefix.[3] When using dynamic host configuration protocol for IPv6 (DHCPv6) [19], a host usually has one global unicast address within a /64 IPv6 prefix. However, even if we assume that there are 1,000,000 hosts and each has three global unicast addresses within a /64 prefix, the address space usage for the /64 prefix is only approximately $1.63 * 10^{-13}$, a number little greater than zero. Apparently, we do not take advantage of the entire IPv6 address space.

2) Multiple Addresses per Host: As mentioned above, a host can generally obtain two SLAAC addresses and one DHCPv6 address within a prefix. One of the SLAAC addresses varies over time (i.e., temporary address) and is used for outgoing connections. The other does not vary over time (i.e., stable address) and is used for incoming connections. However, even if we assume that every host is assigned 100 addresses, the space usage for a prefix is still about $5.42 * 10^{-12}$. Moreover, RFC 7934 [20] also points out that many host functions require more than one IP address to be available. In short, current IPv6 address usage practices remind us that hosts can be

[3]Although this is not mentioned in [18], it is common practice for most current operating systems to configure two SLAAC addresses for hosts.

assigned more addresses to achieve specific goals, e.g., privacy protection.

3) Standardized and Well-Maintained Protocols: For a long time, protocols related to address allocation and security have been created and maintained by the IETF, including DHCPv6 [19], SAVI [21], IPsec [22], and RPKI [23]. Using these protocols instead of creating new, similar ones can save us a lot of effort. Moreover, while creating a new protocol, we often encounter problems similar to those faced when working with existing protocols. For example, APNA [10] encounters the similar certificate management issue, as does IPsec.

C. Design Goals

1) Accountability: Accountability in general means being called to account for one's actions to some authority [24]. At the network layer in computing, accountability refers to the characteristic whereby all data packets can be attributed to their senders [9], [10]. Achieving accountability in the Internet involves a number of procedures. First, the authenticity of data packets should be ensured because they are used in the network to account for misbehaving senders, especially the information in the packet headers. Second, every sender should have a unique identifier in the network, and must be authenticated before they access the network. Third, each data packet should be associated with a specific sender identifier. Many useful protocols can also be designed to use accountability to achieve specific goals, e.g., blocking and filtering traffic, and scoring each sender's reputation.

2) Privacy: Considering the widely accepted definition of privacy at the network layer [9], [10], [25], we have four privacy goals as shown in Table I.

- **Sender Anonymity.** The first is sender anonymity, which provides the capability for a sender to hide its identity from third-party observers in the source AS, transit ASes, destination AS, and even the receiver. We do not consider sender anonymity to be against observers in the LAN segment because they already know the identity (link layer address) of the sender. Nor do we consider sender anonymity to be against the source AS because it knows the identity and network attachment of the sender.
- **Sender-Flow Unlinkability.** Our second privacy goal is sender-flow unlinkability, which means that an adversary cannot obtain more information about the source relativity of flows by observing any number of flows originating from the same AS. The source relativity of flows refers to the possibility of two flows originating from the same sender. Note that the meaning of flow here is the same as the meaning of flow for the sender and receiver in the traditional networks but different for network devices and observers in the network. This is because PAVI changes the source or destination addresses during the transmission process of data packets.
- **Sender-Receiver Unlinkability.** The third goal is sender-receiver unlinkability. No observer can obtain more information about whether the sender communicates with the same receiver in different flows. Note that for those who know the identity of one of the sender and the receiver, achieving sender-receiver unlinkability is to ensure the privacy of the identity of the other.
- **Data Confidentiality.** The fourth goal of privacy here is data confidentiality. Only the receiver can know the packet payload, and all other entities, including the source AS, should know nothing about it. In this paper, we delegate the privacy of packet payload to well-established protocols, e.g., IPsec [22].

Note that privacy leakage from higher layers is beyond the scope of this paper.

3) Deployability: Compared with prevalent protocols [9], [10], we aim to make PAVI more deployable, which implies two sub-goals:

- **Lightweight Enhancements**: PAVI should introduce only lightweight enhancements to the Internet, and should not require new communication identifiers and large-scale modifications to fully deployed infrastructures and protocols, unlike [9], [10]. PAVI should only add a slight performance overhead to the deployed Internet base.
- **Incrementally Deployable**: We aim to make PAVI incrementally deployable. Upgraded ASes can benefit from a partially deployed PAVI immediately whereas legacy ASes can continue operating.

D. Adversary Model

The goal of the adversary against the accountability of PAVI is to create packets associated with someone else in the network without being identified, whereas the goal against the privacy of PAVI is to determine the identity of a sender, whether two flows from the same source AS originate from the same sender, whether two flows have the same sender and receiver identities, or the packet payload. We assume that the adversary can observe packet headers and payload. It is assumed that the hosts can be compromised and used to launch attacks. We also assumed that transit ASes may be malicious. Note that secret keys used in PAVI are well protected, and the adversary cannot compromise them.

III. Design Overview

In this section, we describe a high-level overview of PAVI. The protocol details are provided in Section IV. First of all, we introduce the notion of accountable privacy addresses (APAs) that can be used to associate with users in a cryptographic and stateless manner. Note that, in essence, APAs are short-lived IPv6 addresses with user identities embedded. They can be used to account for misbehaving users and protect user privacy simultaneously. By leveraging IPv6 addresses, PAVI does not introduce new communication identifiers and avoids large-scale modifications to the deployed Internet base.

In each AS, many entities join the communication.

- **Users** use hosts to access the Internet.
- **Hosts** send or receive packets to or from the Internet.
- **Registry Manager** authenticates users and shares secret symmetric keys with users.
- **APA Assigner** generates APAs and assigns them to hosts.
- **Source Monitor** observes APA assignment procedures, creates binding entries between APAs and link-layer

TABLE I
PRIVACY GOALS IN PAVI

Observers	Sender Anonymity	Sender-Flow Unlinkability	Sender-Receiver Unlinkability	Data Confidentiality
Observers in source LAN segment	✘	✘	✔	✔
Observers in source AS	✔	✔	✔	✔
Source AS	✘	✘	✔	✔
Transit ASes	✔	✔	✔	✔
Destination AS	✔	✔	✔	✔
Observers in destination AS	✔	✔	✔	✔
Observers in destination LAN segment	✔	✔	✔	✔
Receiver	✔	✔	✔	✘

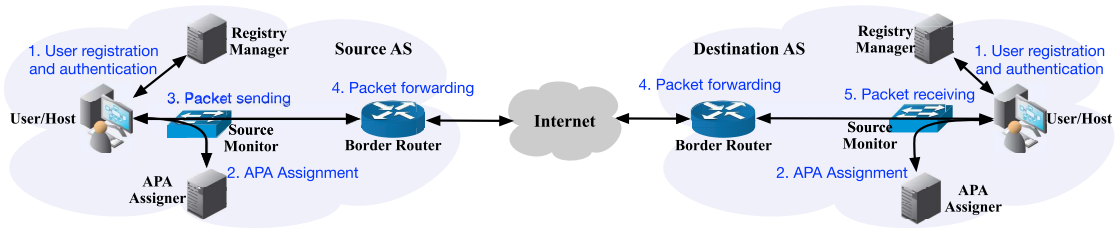


Fig. 1. High-level overview of PAVI.

properties of host's network attachments, and filters source spoofing traffic using binding entries.

- **Border Router** masks the source and destination addresses of packets and forwards packet flows.

As shown in Fig. 1, the communication between hosts consists of five steps:

1) **User Registration and Authentication:** A user subscribes to a preferred privacy service at its ISP, authenticates to its ISP, and shares a symmetric key with it securely during authentication (see Section IV-B).
2) **APA Assignment:** The host in which the user resides sends APA requests to the APA Assigner and configures APAs embedded with his/her UID (see Section IV-C).
3) **Packet Sending:** The source host sends packet flows each with a randomly chosen APA as source address and a masked destination address. It also provides integrity protection to each packet by computing a cryptographic code over some fields of the packet (see Section IV-D).
4) **Packet Forwarding:** The border routers of the source and destination AS mask the destination and source addresses, respectively, to increase sender-receiver unlinkability (see Section IV-E).
5) **Packet Receiving:** The destination host unmasks the source address of each packet to communicate with the source host (see Section IV-F).

IV. DETAILS OF PAVI

This section presents details of the design of PAVI, beginning with user registration and authentication, and followed by APA assignment, packet sending, packet forwarding, and packet receiving. The notation we use while describing the protocol is summarized in Table II.

A. Assumptions

- *All entities can verify prefixes to which an AS can originate routes.* For example, a scheme such as RPKI [23] can be used by legitimate holders to authorize ASes to originate routes to one or more prefixes.

TABLE II
SUMMARY OF SYMBOLS AND NOTATION

Symbol	Remarks
$\mathcal{U}_i$	User i
$\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$	The source host used by $\mathcal{U}_i$
$\mathcal{H}_{\mathcal{D}}^{\mathcal{U}_j}$	The destination host used by $\mathcal{U}_j$
$\boldsymbol{A}^{\mathcal{U}_i}$	A stable IPv6 address assigned to $\mathcal{U}_i$
$\mathsf{A}^{\mathcal{U}_i}$	A set of APAs assigned to $\mathcal{U}_i$
$\mathsf{A}_{\mathcal{S}}$	The original source address
$\mathsf{A}_{\mathcal{D}}$	The original destination address
UID_i	User identifier (UID) assigned to $\mathcal{U}_i$
N_i	The maximum number of APAs assigned to $\mathcal{U}_i$
$k_{\mathcal{A}_i}$	Secret key of AS $\mathcal{A}_i$
$k_{\mathcal{A}_i\mathcal{H}_j}$	Secret key shared between AS $\mathcal{A}_i$ and host $\mathcal{H}_j$
$k_{\mathcal{A}_i\mathcal{A}_j}$	Secret key shared between AS $\mathcal{A}_i$ and AS $\mathcal{A}_j$
$E_k(\cdot)$	Symmetric encryption using key k
$E_k^{-1}(\cdot)$	Symmetric decryption using key k
$MAC_k(\cdot)$	Message authentication code (MAC) using key k

- *Each AS shares secret symmetric keys with other ASes.* After obtaining public keys from the RPKI, the ASes can use a non-interactive Diffie-Hellman key exchange [26], [27] to establish secret symmetric keys.
- *We assume that the cryptographic mechanisms are secure.* That is, encryptions cannot be violated and MACs cannot be forged.

B. User Registration and Authentication

PAVI aims to provide privacy to users. Internet service providers (ISPs) have business incentives to provide privacy to customers. In PAVI, ISPs are allowed to provide different levels of privacy to their customers. It is assumed that an ISP has M users $\{\mathcal{U}_1, \mathcal{U}_2, \cdots, \mathcal{U}_M\}$. When $\mathcal{U}_i$ subscribes to the ISP, the ISP creates a unique user identifier UID_i and the corresponding authentication credentials for him/her. UID_i is a number used by the ISP to identify a user, and it is not required to be self-certifying, unlike HIDs in APIP. $\mathcal{U}_i$ can also purchase different levels of privacy by choosing the preferred number N_i of APAs that the ISP assigns to him/her.

As PAVI associates users with packets by embedding UIDs into APAs, it is necessary to perform user authentication before APA assignment. The Registry Manager uses a layer-2 authentication mechanism to authenticate users. Several well-designed authentication mechanisms can be used, e.g., the RADIUS protocol [28], the Diameter base protocol [29]. During authentication, the host creates a secret symmetric key $k_{\mathcal{A}_S\mathcal{H}_S}$ and securely shares it with the Registry Manager, which maintains an authenticated user list and keeps updating it with other entities in the AS, including the APA Assigner and border routers. The Registry Manager also sends (UID_i, $k_{\mathcal{A}_S\mathcal{H}_S}$) to border routers.

C. APA Assignment

The goal of the APA assignment procedure is to assign addresses to hosts. PAVI retains the current common practice of address assignment used in SLAAC as mentioned in Section II-B.2. More specifically, the APA Assigner assigns one stable IPv6 address and multiple APAs to a host. The stable IPv6 address is used for incoming connections, while the APAs are used for outgoing connections.

Take $\mathcal{U}_i$ for example, the APA Assigner will assign one IPv6 address and N_i APAs embedded with UID_i to the requesting host $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$. Following user authentication, $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ begins to request APAs from the APA Assigner. As the Registry Manager has synchronized the update with the authenticated user list, the APA Assigner generates a stable IPv6 address and N_i APAs $\{\mathsf{A}_1^{\mathcal{U}_i}, \mathsf{A}_2^{\mathcal{U}_i}, \cdots, \mathsf{A}_{N_i}^{\mathcal{U}_i}\}$, and assigns them to $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$. The stable IPv6 address $\boldsymbol{A}^{\mathcal{U}_i}$ is generated as follows:

$$\boldsymbol{A}^{\mathcal{U}_i} = prefix||E_{k_{\mathcal{A}_S}}(UID_i||t||d), \tag{1}$$

where $prefix$ is the IPv6 prefix used to generate $\boldsymbol{A}^{\mathcal{U}_i}$, t is the time at which the APA Assigner generates $\boldsymbol{A}^{\mathcal{U}_i}$, d is a counter employed to resolve duplicate detection and incremented by 1 for each new generation, and $||$ represents concatenation. Similarly, each APA $\mathsf{A}_j^{\mathcal{U}_i}$ is generated as follows:

$$\mathsf{A}_j^{\mathcal{U}_i} = prefix||E_{k_{\mathcal{A}_S}}(UID_i||n_j||t_j||d_j), \tag{2}$$

where $prefix$ is the IPv6 prefix used to generate $\mathsf{A}_j^{\mathcal{U}_i}$, n_j is a counter employed to indicate the number of $\mathsf{A}_j^{\mathcal{U}_i}$, t_j is the time at which the APA Assigner generates $\mathsf{A}_j^{\mathcal{U}_i}$, and d_j is a counter employed to resolve duplicate detection and incremented by 1 for each new generation. Note that these APAs may have different preferred lifetimes. APAs are not self-certifying identifiers but are generated and certified by ISPs.

When an APA expires, $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ should get it into a deprecated state if there are still applications using it, which means it can no longer be used as a source address in new communications. If no connections have been using it, $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ should abort the APA. Then, $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ starts requesting new APAs from APA Assigner.

A source monitor observes the APA assignment procedure and creates binding entries between APAs and a link-layer property of the network attachment of $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ (e.g., switch port). Such mechanisms as [21], [30] can be used to achieve this goal. The source monitor's functionalities are currently widely supported by commercial products, e.g., Cisco [31], Huawei [32], and Juniper [33].

D. Packet Sending

Once a source host $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ obtains $\mathsf{A}^{\mathcal{U}_i} = \{\mathsf{A}_1^{\mathcal{U}_i}, \mathsf{A}_2^{\mathcal{U}_i}, \cdots, \mathsf{A}_{N_i}^{\mathcal{U}_i}\}$ from its AS, it can start communicating with other hosts. If $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ decides to protect data confidentiality, it uses the ESP header [34]. Each packet flow originating from $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ is processed in the following three steps:

*1) **APA Determination**:* Once the peer host has been determined (e.g., through DNS lookup), $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ sends packet flows each with a source APA randomly chosen from $\mathsf{A}^{\mathcal{U}_i}$. Sending flows using different source APAs improves unlinkability between a sender and its flows. The number of packet flows using $\mathsf{A}_j^{\mathcal{U}_i}$ depends on $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$, and one flow per APA is the best choice (See Section V-B.2 for more details).

*2) **PAVI Header Insertion**:* $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ must insert the PAVI header into each packet to protect the integrity of the packets. Details of the PAVI header are as follows:

- Next Header: a one-byte field which identifies the type of the next payload following the PAVI header. A new value should be assigned by the Internet Assigned Numbers Authority to this new type of extension header.
- Payload Length: a two-byte field that specifies the length of the PAVI header in four-byte units. For example, if a MAC algorithm yields a 128-bit authentication value, this field will be "9".
- Reserved: a one-byte field reserved for future use. It must be set to"zero" by the sender and ignored by the receiver. Note that this value should be included in the IPC calculation, but is otherwise ignored by the receiver.
- Extra Encrypted Address (EEA): an eight-byte value which is used by hosts and border routers to store extra encrypted address bits.
- Timestamp (TS): a four-byte value that identifies the time at which the sender inserts the PAVI header into the packet.
- Sequence Number (SN): a four-byte field which contains a counter that increases by one for each packet. The first packet in a flow will have a random sequence number.
- Integrity Protection Code (IPC): a variable-length field that contains the integrity check value for the packet. Note that this value must be an integral multiple of four bytes in length. The IPC is generated as follows:

$$IPC = MAC_{k_{\mathcal{A}_S\mathcal{H}_S}}(IP_{src}||IP_{dst}||EEA||TS \\ ||SN||H(payload)), \tag{3}$$

  where IP_{src} is the source address, IP_{dst} is the destination address, TS is the timestamp, SN is the sequence number, and $H(payload)$ is the hash encryption of the upper layer payload.

The IPC is computed over the packet using the secret symmetric key $k_{\mathcal{A}_S\mathcal{H}_S}$ shared with its AS, which means that the PAVI header is used to improve the accountability of the packet instead of it being verified by the peer host.

*3) **Destination Masking**:* $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ encrypts the interface identifier (IID) of the destination IPv6 address $\mathsf{A}_{\mathcal{D}}$ of each packet using the secret symmetric key $k_{\mathcal{A}_S\mathcal{H}_S}$ as follows:

$$IID_{\mathcal{A}_S\mathcal{H}_S}||EEA_{\mathcal{A}_S\mathcal{H}_S} = E_{k_{\mathcal{A}_S\mathcal{H}_S}}(IID_{\mathcal{D}}||TS||SN), \tag{4}$$

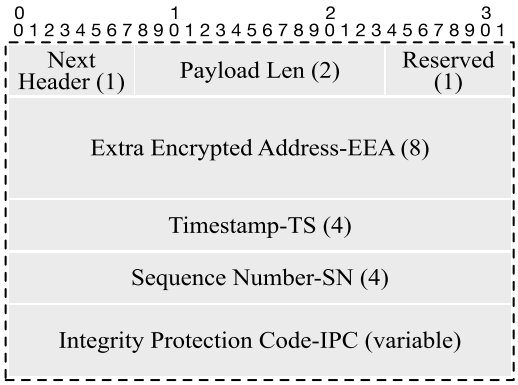


Fig. 2. PAVI header.

where $IID_{\mathcal{D}}$ is the IID of destination address $\mathsf{A}_{\mathcal{D}}$. Then, $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ uses the first 64 bits of the result of encryption as the new IID of the destination address and the last 64 bits as EEA in the PAVI header, and sends the packets. The goal of this step is to improve sender-receiver unlinkability against observers in the same LAN segment as $\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$ (See more details in Section V-B.2).

The source monitor filters packets that the source IPv6 addresses do not match the binding entries. This step is to ensure that no source spoofing packets leave the access network.

E. Packet Forwarding

As PAVI is designed based on IPv6 and only changes the generation mechanism of IID, it does not change the routing of today's Internet, which is different from APIP and APNA.

1) Outgoing Packets: When a border router of the source AS receives an outgoing packet, it first uses the secret key $k_{\mathcal{A}_S\mathcal{H}_S}$ to decrypt the IID of the destination address and EEA in the PAVI header to obtain the original destination address $\mathsf{A}_{\mathcal{D}}$:

$$IID_{\mathcal{D}}||TS||SN = E^{-1}_{k_{\mathcal{A}_S\mathcal{H}_S}}(IID_{\mathcal{A}_S\mathcal{H}_S}||EEA_{\mathcal{A}_S\mathcal{H}_S})). \quad (5)$$

Then it verifies packet integrity by computing an IPC using the IP header and payload, and comparing it with the one in the PAVI header. If the validation fails, the border router drops the packet. Otherwise, it re-encrypts the IID of $\mathsf{A}_{\mathcal{D}}$ using secret key $k_{\mathcal{A}_S\mathcal{A}_D}$ shared with the destination AS and forwards the packet:

$$IID_{\mathcal{A}_S\mathcal{A}_D}||EEA_{\mathcal{A}_S\mathcal{A}_D} = E_{k_{\mathcal{A}_S\mathcal{A}_D}}(IID_{\mathcal{D}}||TS||SN). \quad (6)$$

Algorithm 1 summarizes these steps.

In summary, only authenticated packets can leave the source AS following these checks.

2) Incoming Packets: Algorithm 2 summarizes the processing of incoming packets. More specifically, when a border router of the destination AS receives an incoming packet, it first uses the source prefix to search the secret symmetric key $k_{\mathcal{A}_S\mathcal{A}_D}$ shared with the source AS. Then, it decrypts the IID of the destination address and EEA in the PAVI header to obtain the original destination address $\mathsf{A}_{\mathcal{D}}$:

$$IID_{\mathcal{D}}||TS||SN = E^{-1}_{k_{\mathcal{A}_S\mathcal{A}_D}}(IID_{\mathcal{A}_S\mathcal{A}_D}||EEA_{\mathcal{A}_S\mathcal{A}_D}). \quad (7)$$

Algorithm 1 Processing of Outgoing Packets

Require:
- ▷ pkt refers to the whole packet
- ▷ pkt_hdr refers to packet header
- ▷ $host_keys$ refers to host key table
- ▷ as_keys refers to AS key table
- ▷ $pavi[]$ is the PAVI header that the source creates
- ▷ $k_{\mathcal{A}_S}$ is the secret key of source AS

$src_addr \leftarrow pkt_hdr[IP_{src}]$
$dst_addr \leftarrow pkt_hdr[IP_{dst}]$
$(prefix_s, iid_s) \leftarrow split(src_addr)$
$(prefix_d, iid_d) \leftarrow split(dst_addr)$
$eea \leftarrow pavi[EEA]$
$(uid, n, t, d) \leftarrow E^{-1}_{k_{\mathcal{A}_S}}(iid_s)$
$k_{\mathcal{A}_S\mathcal{H}_S} \leftarrow host_keys[uid]$
$ipc = MAC_{k_{\mathcal{A}_S\mathcal{H}_S}}(src_addr||dst_addr||eea||pavi[TS]$
$||pavi[SN]||H(pkt[payload]))$
if $ipc == pavi[IPC]$ **then**
  $(iid_d, ts, sn) \leftarrow E^{-1}_{k_{\mathcal{A}_S\mathcal{H}_S}}(iid_d||eea)$
  $(iid_d, eea) \leftarrow E_{k_{\mathcal{A}_S\mathcal{A}_D}}(iid_d||ts||sn)$
  $k_{\mathcal{A}_S\mathcal{A}_D} \leftarrow as_keys[prefix_d]$
  $pkt_hdr[IP_{dst}] \leftarrow prefix_d||iid_d$
  $pavi[EEA] \leftarrow eea$
  forward packet
end if
drop packet

Next, it decrypts the IID of $\mathsf{A}_{\mathcal{D}}$ to get the UID of the destination, and looks up the host key table to get the secret symmetric key $k_{\mathcal{A}_D\mathcal{H}_D}$ shared with the destination host $\mathcal{H}_{\mathcal{D}}^{\mathcal{U}_j}$. In Algorithm 2, $\mathsf{A}_{\mathcal{D}}$ is assumed to be a stable address. After encrypting the IID of the original source address $\mathsf{A}_{\mathcal{S}}$ with key $k_{\mathcal{A}_D\mathcal{H}_D}$, it forwards the packet:

$$IID_{\mathcal{A}_D\mathcal{H}_D}||EEA_{\mathcal{A}_D\mathcal{H}_D} = E_{k_{\mathcal{A}_D\mathcal{H}_D}}(IID_{\mathcal{S}}||TS||SN). \quad (8)$$

Note that if the sender uses each APA to send only one flow, the encryption of the original source address can be ignored.

F. Packet Receiving

When $\mathcal{H}_{\mathcal{D}}^{\mathcal{U}_j}$ receives the packet, it decrypts the IID of the source address to determine the real source address $\mathsf{A}_{\mathcal{S}}$ of the host with which it communicates:

$$IID_{\mathcal{S}}||TS||SN = E^{-1}_{k_{\mathcal{A}_D\mathcal{H}_D}}(IID_{\mathcal{A}_D\mathcal{H}_D}||EEA_{\mathcal{A}_D\mathcal{H}_D}). \quad (9)$$

Once $\mathcal{H}_{\mathcal{D}}^{\mathcal{U}_j}$ assumes that packet flows from the sender are malicious or unwanted, $\mathcal{H}_{\mathcal{D}}^{\mathcal{U}_j}$ can require its AS $\mathcal{A}_D$ to block traffic. When $\mathcal{A}_D$ receives the block request, it sends the request to $\mathcal{A}_S$ with the signature signed by its private key. $\mathcal{A}_S$ blocks traffic from the specific APA or user if the block request is valid, such as through Shutoff [9], [10].

V. Protocol Analysis

In this section, we show how PAVI prevents attacks that undermine accountability and privacy, and quantitatively analyze accountability and privacy.

Algorithm 2 Processing of Incoming Packets

Require:

▷ pkt_hdr refers to packet header
▷ $host_keys$ refers to host key table
▷ as_keys refers to AS key table
▷ $pavi[]$ is the PAVI header that the source creates
▷ $k_{\mathcal{A}_\mathcal{D}}$ is the secret key of destination AS

$src_addr \leftarrow pkt_hdr[IP_{src}]$
$dst_addr \leftarrow pkt_hdr[IP_{dst}]$
$(prefix_s, iid_s) \leftarrow split(src_addr)$
$(prefix_d, iid_d) \leftarrow split(dst_addr)$
$eea \leftarrow pavi[EEA]$
$k_{\mathcal{A}_S\mathcal{A}_D} \leftarrow as_keys[prefix_s]$
$(iid_d, ts, sn) \leftarrow E^{-1}_{k_{\mathcal{A}_S\mathcal{A}_D}}(iid_d||eea)$
$(uid, t, d) \leftarrow E^{-1}_{k_{\mathcal{A}_\mathcal{D}}}(iid_d)$
$k_{\mathcal{A}_D\mathcal{H}_D} \leftarrow host_keys[uid]$
$(iid_s, eea) \leftarrow E_{k_{\mathcal{A}_D\mathcal{H}_D}}(iid_s||ts||sn)$
$pkt_hdr[IP_{src}] \leftarrow prefix_s||iid_s$
$pkt_hdr[IP_{dst}] \leftarrow prefix_d||iid_d$
$pavi[EEA] \leftarrow eea$
forward packet

A. Security Analysis

1) Spoofing APAs: The adversary may attempt to spoof APAs in two ways. The first way is that he may generate new IPv6 addresses randomly by himself and use them as APAs. The second way is that he uses APAs belonging to other hosts. However, packets with spoofing APAs are dropped because the source monitor validates the source address of each packet according to its binding entries.

Here, the discussion is based on the assumption that all hosts are directly connected to source monitors with each host's APAs associated with different link-layer properties. If source monitors are located far from hosts, these hosts under the same source monitor can spoof each other's APAs to communicate with peers. Therefore, the nearer to hosts source monitors are deployed, the better the performance of source monitors on preventing spoofing APAs is.

2) Integrity Attacks: A malicious host may send packets with incorrect IPCs. In PAVI, a border router first decrypts IID of the destination address and EEA in the PAVI header to obtain original destination address $A_\mathcal{D}$. Then, it encrypts source address, destination address, EEA, TS, SN, and the hash of payload as mentioned in Equation 3 and gets a new IPC′. Finally, the border router compares IPC′ with IPC in the PAVI header to determine whether the packet is bogus or not. If they are not the same, the border router drops the packets. Therefore, packets that do not pass the IPC check will not be forwarded to the Internet.

3) Identity Minting: Identity minting is when a malicious user attempts to create and use unauthorized UIDs to connect to the Internet. In PAVI, every user obtains authentication credentials used to access the Internet when subscribing to an ISP. PAVI allows an ISP to provide only a unique UID to every user within it. This means that an adversary cannot succeed in the user authentication process if he does not have the corresponding authentication credentials of another user. Similarly, if the adversary uses unauthorized UIDs to authenticate to the ISP, unauthorized UIDs will fail the user authentication process because the ISP cannot find legal entries of these UIDs.

B. Quantitative Analysis

1) Accountability Analysis: We choose an AS as a unit to discuss accountability. The goal of accountability at the network layer is to determine the true user identities of all the data packets originating from an AS. The accountability of an AS $\mathscr{S}$ can be defined as:

$$A_\mathscr{S}(P) = \frac{\sum_{p\in P} a(p)}{||P||}, \tag{10}$$

where P is a set containing all data packets originating from $\mathscr{S}$, p is a packet in P, $||P||$ is the total number of packets in P, and $a(p)$ is the accountability estimation for p. Note that $a(p)$ is a value between zero and one.

As mentioned in Section II-C.1, the core idea of using a packet for accounting for a sender is to ensure the packet's authenticity first and then extract a sender identifier from the packet to identify the sender. The more senders the sender identifier is mapped to, the less accountability the system has. Therefore, we can use several related definitions to compute $a(p)$:

- $t(p)$, a Boolean value, is the authenticity of packet p. More specifically, the packet is "true" when $t(p) = 1$ and "not true" when $t(p) = 0$.
- $r(p)$ is used to extract the sender identifier from p.
- $u_T(id)$ is the number of senders that use id during period T.

Then, $a(p)$ is defined as follows:

$$a(p) = \frac{t(p)}{u_T(r(p))}. \tag{11}$$

We then compute the accountability of several protocols under the worst-case scenario, as shown in Table III. N denotes the number of hosts mapping to the same IP address during period T using NAT, and M denotes the number of hosts that use the same IP address during period T in Persona. For APIP, only the first packet is authentic and vouched for by the host's accountability delegate because "whitelisted" hosts can ignore sending packet fingerprints to their delegates. Therefore, only a few packets can be accounted for during the verification cycle. V is the collection of the first packets verified along the forwarding path during one verification cycle. In this case, P in Equation (10) means the set of packets in one verification interval. As the APNA embeds host identifiers into their newly defined addresses (i.e., EphIDs) and uses MACs to protect the authenticity and integrity of packets, it has good accountability. PAVI adopts well-designed monitoring techniques to prevent source spoofing, uses IPCs to protect the integrity of packets, and associates users with their packets via APAs, which also yields satisfactory accountability.

2) Privacy Analysis: We use information-theoretic entropy [35] as the measure of information contained in a probability distribution to analyze sender anonymity, sender-flow unlinkability, and sender-receiver unlinkability. Recall that there are M users $\mathcal{U} = \{\mathcal{U}_1, \mathcal{U}_2, \cdots, \mathcal{U}_M\}$ in an AS. $\mathcal{U}_i$ has N_i APAs

TABLE III
COMPARISON OF ACCOUNTABILITY

	$t(p)$	$r(p)$	$u_T(id)$	$a(p)$	$A_{\mathscr{S}}(P)$
Today's Internet	0	IP	N/A	0	0
NAT (True source)	1	IP	N	$\frac{1}{N}$	$\frac{1}{N}$
Persona	1	IP	M	$\frac{1}{M}$	$\frac{1}{M}$
APIP	1/0	Public key	1	1/0	$\frac{\|\|V\|\|}{\|\|P\|\|}$
APNA	1	HID	1	1	1
PAVI	1	UID	1	1	1

$\mathsf{A}^{\mathcal{U}_i} = \{\mathsf{A}_1^{\mathcal{U}_i}, \mathsf{A}_2^{\mathcal{U}_i}, \cdots, \mathsf{A}_{N_i}^{\mathcal{U}_i}\}$ and uses each APA to send flows. Therefore, we have $\mathsf{A} = \{\mathsf{A}_j^{\mathcal{U}_i} | 1 \leq i \leq M, 1 \leq j \leq N_i\}$.

Sender Anonymity. We measure the possible information leakage of sender identity based on degree of anonymity [36]. For example, an adversary $\mathscr{T}$ detects many APAs by observing traffic originating from an AS. The goal of the adversary is to extract the user identity of each APA. For each APA, it maintains a probability distribution X. Specifically, the adversary $\mathscr{T}$ believes that it belongs to user $\mathcal{U}_i$ with probability $p_i := P_{\mathscr{T}}(X = \mathcal{U}_i)$. Then, the entropy is defined as:

$$H_{\mathscr{T}}(X) = -\sum_{i=1}^{M} p_i \cdot log_2(p_i), \tag{12}$$

where $\sum_{i=1}^{M} p_i = 1$. The maximum entropy is $max(H_{\mathscr{T}}(X)) = log_2(M)$ when $p_i = \frac{1}{M}$. In PAVI, as $\mathscr{T}$ does not know the secret key of the AS, it cannot decrypt the APA to obtain the user identifier. Accordingly, the adversary believes that the APA can belong to any user in the AS with probability $p_i = \frac{1}{M}$. Therefore, we can calculate entropy:

$$H_{\mathscr{T}} = -\sum_{i=1}^{M} \frac{1}{M} \cdot log_2(\frac{1}{M}) = log_2(M). \tag{13}$$

Based on [36], the degree of anonymity after introducing PAVI is:

$$d = 1 - \frac{max(H_{\mathscr{T}}(X)) - H_{\mathscr{T}}}{max(H_{\mathscr{T}}(X))} = \frac{H_{\mathscr{T}}}{max(H_{\mathscr{T}}(X))} = 1. \tag{14}$$

Although each APA contains a user identifier, the adversary cannot obtain the real user identifier from it due to encryption protection. However, the IPv6 Internet uses IPv6 addresses to identify hosts for quite a long time, which makes it easy for $\mathscr{T}$ to determine the host's identity and obtain $d = 0$.

Sender-Flow Unlinkability. We also use entropy to measure sender-flow unlinkability. In PAVI, an adversary $\mathscr{T}$ observes outgoing traffic originating from the AS and detects $N = \sum_{i=1}^{M} N_i$ different source APAs. If $\mathscr{T}$ knows nothing about users in the AS, his goal is to partition the APA set A into a set of non-empty, pairwise disjoint subsets. According to [37], the unlinkability of the APAs in A for $\mathcal{U}$ against $\mathscr{T}$ is defined as:

$$U_{\mathsf{A}}(\mathscr{T}_{\emptyset}, \mathcal{U}) = log_2(B(N)), \tag{15}$$

where $\mathscr{T}_{\emptyset}$ denotes the adversary without any hint and $B(N)$ is the total number of partitions of A, a Bell number [38], [39], which can be calculated by the recursive formula:

$$B(0) = 1,$$

$$B(N) = \sum_{k=0}^{N-1} \binom{N-1}{k} B(k). \tag{16}$$

However, if $\mathscr{T}$ knows a hint H whereby M users in the AS send flows, the goal of $\mathscr{T}$ is to partition set A into M non-empty, pairwise disjoint subsets. Then, the unlinkability of the APAs in A for $\mathcal{U}$ against $\mathscr{T}$ is:

$$U_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U}) = log_2(S(N, M)), \tag{17}$$

where $\mathscr{T}_{\mathsf{H}}$ denotes the adversary with hint H and $S(N, M)$ is a Stirling number of the second kind [40], which can be calculated by the formula:

$$S(N, M) = \frac{1}{M!} \sum_{k=0}^{M} (-1)^k \binom{M}{k} (M-k)^N. \tag{18}$$

Therefore, the degree of unlinkability of the APAs in A for $\mathcal{U}$ against $\mathscr{T}$ with access to hint H is:

$$D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U}) = \frac{U_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})}{U_{\mathsf{A}}(\mathscr{T}_{\emptyset}, \mathcal{U})} = \frac{log_2(S(N, M))}{log_2(B(N))}. \tag{19}$$

Fig. 3(a) and Fig. 3(b) show the variation in the degree of sender-flow unlinkability $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})$ with respect to the number of users (M) and the number of APAs (N_i) assigned to each user in an AS, respectively. We see that $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})$ increases with M, which means senders and flows become more unlinkable as the number of users increases. This result is as expected because more users and more APAs will make it more difficult for the adversary $\mathscr{T}$ to link senders to APAs. However, as N_i increases, $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})$ decreases slightly, which means senders and flows will become less unlinkable when assigning each user more APAs.

In another case, if $\mathscr{T}$ knows another hint $\hat{\mathsf{H}}$ whereby $\mathcal{U}_i$ has N_i APAs, the goal of $\mathscr{T}$ is to partition A into a set of non-empty, pairwise disjoint subsets in which there exists at least a subset with N_i APAs. We assume that the number of such partitions is $Q(N, N_i)$. Then, the unlinkability of the APAs in A for $\mathcal{U}_i$ against $\mathscr{T}$ is:

$$U_{\mathsf{A}}(\mathscr{T}_{\hat{\mathsf{H}}}, \mathcal{U}_i) = log_2(Q(N, N_i)), \tag{20}$$

where $\mathscr{T}_{\hat{\mathsf{H}}}$ denotes the adversary with hint $\hat{\mathsf{H}}$. As for $Q(N, N_i)$, we use two steps to calculate it. The first step is to partition A into k subsets and find the number of candidate partitions $R(N, k, N_i, j)$ where there at least exist j subsets with N_i APAs. The result is as follows:

$$R(N, k, N_i, j) = \frac{P(N, N - j \cdot N_i)}{j! \cdot (N_i!)^j} \cdot S(N - j \cdot N_i, k - j), \tag{21}$$

where $P(N, N - j * N_i)$ denotes the number of $(N - j \cdot N_i)$-permutations of N and $S(N - j \cdot N_i, k - j)$ is the number of ways to partition a set with $N - j \cdot N_i$ APAs into $k - j$ subsets. The second step is to add up the number of candidate partitions in each case of k and j by using the inclusion–exclusion principle:

$$Q(N, N_i) = \sum_{k=1}^{N} \sum_{j=1}^{min(k, |N/N_i|)} (-1)^{j+1} \cdot R(N, k, N_i, j), \tag{22}$$

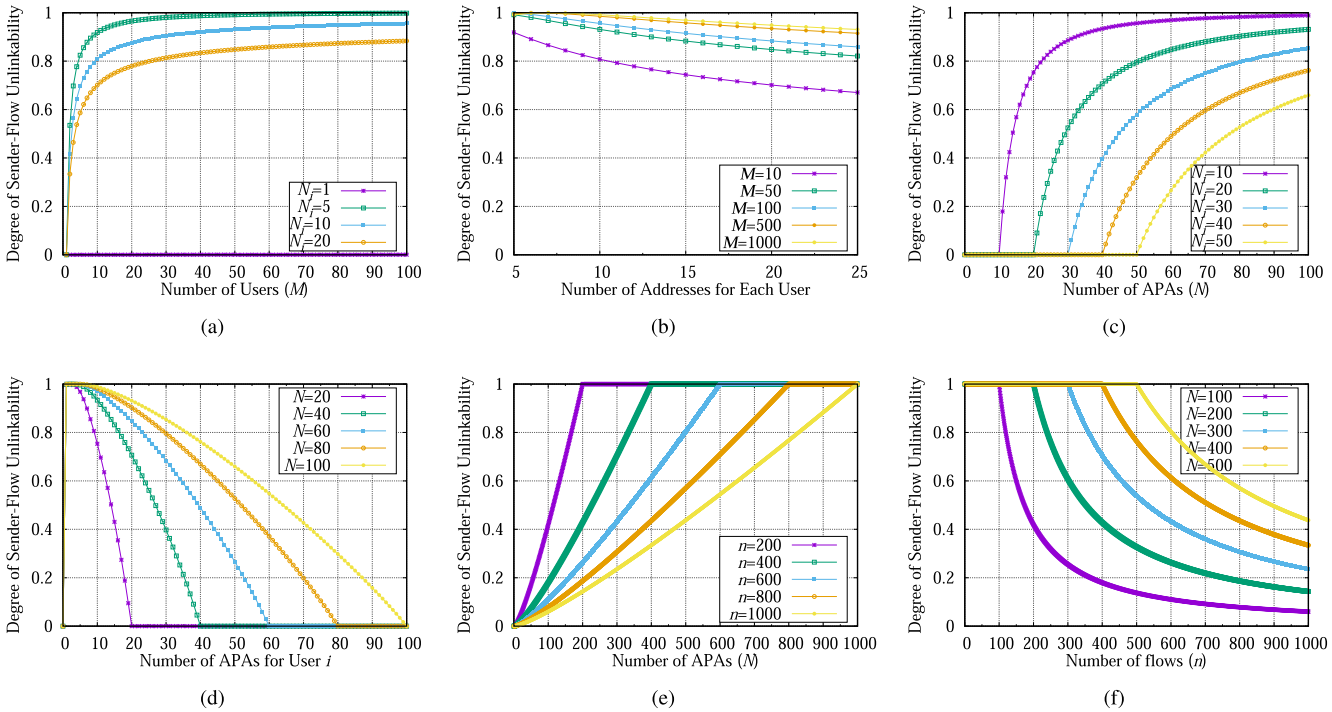


Fig. 3. The degree of sender-flow unlinkability in different scenarios. (a) and (b) show the variation in the degree of sender-flow unlinkability with respect to the number of users (M) and the number of APAs (N_i) assigned to each user in an AS, respectively, when the adversary knows the number of users in the AS. (c) and (d) show the variation in the degree of sender-flow unlinkability with respect to the number of APAs (N) observed by the adversary in an AS and the number of APAs for $\mathcal{U}_i$ (N_i), respectively, when the adversary knows $\mathcal{U}_i$ has N_i APAs. (e) and (f) show that the variation in the degree of sender-flow unlinkability with respect to the number of APAs (N) and the number of flows (n) originating from an AS, respectively.

where $|N/N_i|$ denotes the value of N/N_i rounded down to the nearest integer and $min(k, |N/N_i|)$ returns the smaller one of the two numbers. Similarly, the degree of unlinkability of the APAs in A for $\mathcal{U}_i$ against $\mathscr{T}$ with access to hint $\hat{\mathsf{H}}$ is:

$$D_{\mathsf{A}}(\mathscr{T}_{\hat{\mathsf{H}}}, \mathcal{U}_i) = \frac{U_{\mathsf{A}}(\mathscr{T}_{\hat{\mathsf{H}}}, \mathcal{U}_i)}{U_{\mathsf{A}}(\mathscr{T}_{\emptyset}, \mathcal{U})} = \frac{log_2(Q(N, N_i))}{log_2(B(N))}. \quad (23)$$

Fig. 3(c) and Fig. 3(d) show the variation in the degree of sender-flow unlinkability $D_{\mathsf{A}}(\mathscr{T}_{\hat{\mathsf{H}}}, \mathcal{U})$ with respect to the number of APAs (N) observed by $\mathscr{T}_{\hat{\mathsf{H}}}$ in an AS and the number of APAs for $\mathcal{U}_i$ (N_i), respectively. As we can see, $D_{\mathsf{A}}(\mathscr{T}_{\hat{\mathsf{H}}}, \mathcal{U})$ increases quickly as N increases when the number of APAs for $\mathcal{U}_i$ stays the same. However, $D_{\mathsf{A}}(\mathscr{T}_{\hat{\mathsf{H}}}, \mathcal{U})$ decreases quickly as N_i increases when the total number of APAs N stays the same. This is intuitive because the higher the ratio of APAs for $\mathcal{U}_i$ is, the less candidate partitions which $\mathscr{T}_{\hat{\mathsf{H}}}$ randomly selects from.

An AS can determine the minimum degree of sender-flow unlinkability, and we suggest that an AS should provide degree $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U}) \geq 0.8$. Nevertheless, the minimum acceptable degree for an AS may depend on the unlinkability requirement for the AS, and we believe that such a minimum cannot be suggested before tests.

The above analysis of sender-flow unlinkability is based on APA. Next, we explore the influence of the number of flows originating from an APA on sender-flow unlinkability. We assume that the adversary $\mathscr{T}$ totally detects n flows, denoted as a flow set F, originating from the AS. If all flows have different source APAs, the unlinkability of the flows in F for $\mathcal{U}$ against $\mathscr{T}$ is:

$$U_{\mathsf{F}}(\mathscr{T}_{\emptyset}, \mathcal{U}) = log_2(B(n)). \quad (24)$$

However, each user may use one APA to send more than one flow. We assume that $\mathcal{U}_i$ uses each APA $\mathsf{A}_j^{\mathcal{U}_i}$ to send n_{ij} flows during the adversary's observation time. Thus $\mathcal{U}_i$ totally sends $n_i = \sum_{j=1}^{N_i} n_{ij}$ flows, and $n = \sum_{i=1}^{M} n_i$. $\mathscr{T}$ believes flows with the same source APA belong to the same user, which can be considered as a hint $\tilde{\mathsf{H}}$. and his goal is to partition the APA set A into a set of non-empty, pairwise disjoint subsets whose union is A. In such a case, the degree of unlinkability of the flows in F for $\mathcal{U}$ against $\mathscr{T}$ with access to hint $\tilde{\mathsf{H}}$ is:

$$D_{\mathsf{F}}(\mathscr{T}_{\tilde{\mathsf{H}}}, \mathcal{U}) = \frac{U_{\mathsf{F}}(\mathscr{T}_{\tilde{\mathsf{H}}}, \mathcal{U})}{U_{\mathsf{F}}(\mathscr{T}_{\emptyset}, \mathcal{U})} = \frac{log_2(B(N))}{log_2(B(n))}. \quad (25)$$

Fig. 3(e) and Fig. 3(f) show that the variation in the degree of sender-flow unlinkability $D_{\mathsf{F}}(\mathscr{T}_{\tilde{\mathsf{H}}}, \mathcal{U})$ with respect to the number of APAs (N) and the number of flows (n) observed by $\mathscr{T}_{\tilde{\mathsf{H}}}$ originating from an AS, respectively. Obviously, $D_{\mathsf{F}}(\mathscr{T}_{\tilde{\mathsf{H}}}, \mathcal{U})$ increases as the number of APAs increases when the number of flows stays the same. However, $D_{\mathsf{F}}(\mathscr{T}_{\tilde{\mathsf{H}}}, \mathcal{U})$ decreases quickly as the number of flows increases when the number of APAs stays the same. This means the smaller the number of flows that a host uses an APA to send is, the higher sender-flow unlinkability the host can obtain.

Sender-Receiver Unlinkability. Similar to the measurement of sender-flow unlinkability, we can also measure sender-receiver unlinkability based on the degree of unlinkability. The hints related to the sender, receiver, and their ASes may help the adversary make decisions. For example,

TABLE IV
COMPARISON OF SENDER-RECEIVER UNLINKABILITY IN DIFFERENT SCENARIOS

	H_R^1	H_R^2	H_R^3
H_S^1	1	$\frac{64+log_2 L}{128}$	$\frac{1}{2}$
H_S^2	$\frac{64+log_2 M}{128}$	$\frac{log_2 M+log_2 L}{128}$	$\frac{log_2 M}{128}$
H_S^3	$\frac{1}{2}$	$\frac{log_2 L}{128}$	0

an adversary may compromise hosts residing in the same LAN segment as the sender and obtain hint H_S concerning sender identity that leads to a decline in sender-receiver unlinkability. When a sender sends packets, compromised hosts in the LAN segment can learn the source and destination addresses. In PAVI, however, as the sender encrypts the IID of the destination address in each packet, compromised hosts cannot know the true destination address. The adversary can only guess one of the addresses within the destination prefix with probability $p_i = \frac{1}{2^{64}}$. The degree of unlinkability between sender and receiver against $\mathscr{T}$ with access to hint H_S is defined as:

$$D(\mathscr{T}_{\mathsf{H}_S}) = \frac{U(\mathscr{T}_{\mathsf{H}_S})}{U(\mathscr{T}_{\emptyset})} = \frac{-\sum_{i=1}^{2^{64}} \frac{1}{2^{64}} \cdot log_2(\frac{1}{2^{64}})}{-\sum_{i=1}^{2^{128}} \frac{1}{2^{128}} \cdot log_2(\frac{1}{2^{128}})} = \frac{1}{2}. \quad (26)$$

Moreover, if the adversary $\mathscr{T}$ also knows hint H_R whereby L users are in the destination AS $\mathcal{A}_{\mathcal{D}}$, the degree of unlinkability decreases to:

$$D(\mathscr{T}_{\mathsf{H}_S\mathsf{H}_R}) = \frac{U(\mathscr{T}_{\mathsf{H}_S\mathsf{H}_R})}{U(\mathscr{T}_{\emptyset})} = \frac{-\sum_{i=1}^{L} \frac{1}{L} \cdot log_2(\frac{1}{L})}{-\sum_{i=1}^{2^{128}} \frac{1}{2^{128}} \cdot log_2(\frac{1}{2^{128}})} = \frac{log_2 L}{128}. \quad (27)$$

According to the knowledge of the adversary, we calculate the degree of sender-receiver unlinkability in different scenarios. The results are shown in Table IV. The first row refers to knowledge of the receiver and its AS, and the first column indicates knowledge of the sender and its AS. We consider three typical hints concerning the sender/receiver:

- $\mathsf{H}_S^1/\mathsf{H}_R^1$: the adversary knows nothing about the sender/receiver and its AS.
- $\mathsf{H}_S^2/\mathsf{H}_R^2$: the adversary knows the number (M/L) of users in $\mathcal{A}_{\mathcal{S}}/\mathcal{A}_{\mathcal{D}}$.
- $\mathsf{H}_S^3/\mathsf{H}_R^3$: the adversary knows the sender/receiver identity.

VI. IMPLEMENTATION AND EVALUATION

In this section, we describe the implementation details and performance evaluation of PAVI.

A. Implementation

For source monitors, we used H3C E552 switches that implemented source address validation improvement (SAVI) [21] to create binding entries between APAs and the switch port which a host is connected to and use them to filter source spoofing traffic.

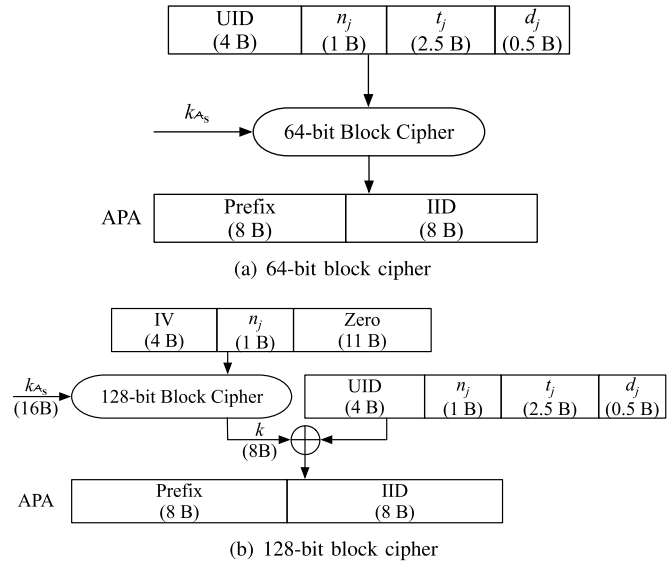


Fig. 4. APA generation procedure.

1) APA Generation: As UID is only a number assigned by an ISP to identify its users and is not required to be self-certifying, we do not specify how the ISP generates and maps UIDs. In our implementation, we assume the ISP uses 32-bit UIDs to identify its users. We obtained a concatenation of a 32-bit UID, an 8-bit indicating counter, a 20-bit timestamp, and a 4-bit detecting counter. Then, according to different encryption algorithms, the procedures differ. If a 64-bit block cipher is adopted, we only need to use the secret key $k_{\mathcal{A}_S}$ to encrypt the concatenation and append the ciphertext to the prefix to obtain an APA, as shown in Fig. 4(a). If a 128-bit block cipher is used, we can use the block cipher and $k_{\mathcal{A}_S}$ to generate a 64-bit encryption key k, and then use k to encrypt the concatenation to obtain the ciphertext, which is as the IID. The whole process is shown in Fig. 4(b).

2) APA Assignment Implementation: We implemented the authentication process using 802.1X [41] and the Radius protocol [28]. We used DHCPv6 servers and APA generators to implement the APA Assigner. APA generators aim to use UIDs to generate APAs after authentication. Fig. 5 sheds light on details of the APA assignment procedure. We used an Ubuntu server installed with the open-source ISC DHCP server software [42] as the DHCPv6 server, in which we modified the processing logic for the identity association for temporary addresses (IA_TA) options. A DHCPv6 server assigned APAs in IA_TA to hosts once it obtained APAs from the APA generators.

3) Border Router Implementation: We implemented the functionalities of the border routers based on the Intel Data Plane Development Kit [43]. To mask the source and destination addresses, we used the AES algorithm through Intel's AES-NI encryption instruction set [44]. To compute the IPC, we leveraged the cipher block chaining mode with AES as the underlying block cipher.

B. Evaluation

1) APA Generation: The APA generation overhead is inherent in cryptography-based security mechanisms. For

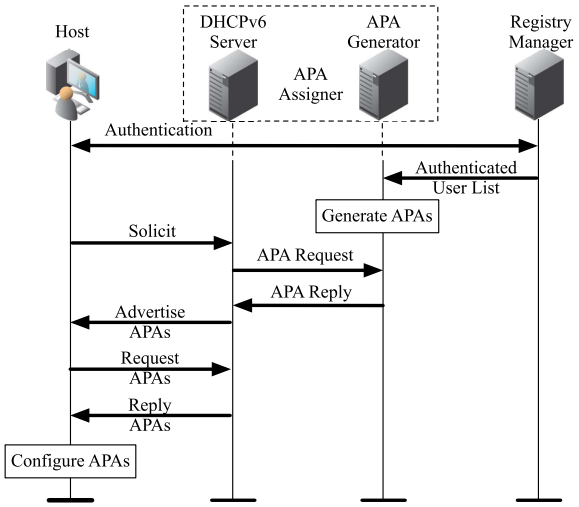


Fig. 5. Implementation of address assignment.

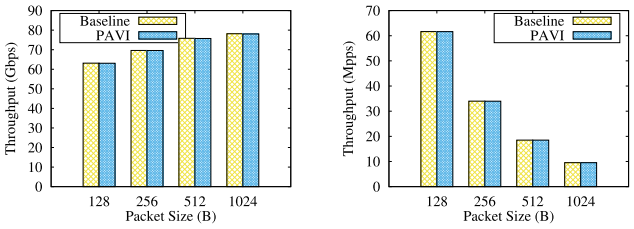
TABLE V
SECURITY LEVEL, ENCRYPTION SIZE, AND COMPUTATIONAL COSTS OF WELL-KNOW ENCRYPTION SCHEMES

Scheme	Security	Block Size	Time
DES	56 bits	64 bits	109 ns
3DES	168 bits	64 bits	270 ns
Blowfish	256 bits	64 bits	66 ns
IDEA	128 bits	64 bits	83 ns
CAST-128	128 bits	64 bits	74 ns
SEED	128 bits	128 bits	99 ns
Camellia-128	128 bits	128 bits	51 ns
AES-128	128 bits	128 bits	15 ns

comparison, we list the APA generation overheads of well-known 64-bit and 128-bit block ciphers in Table V. We used each algorithm to generate 1,000,000 APAs and calculated the average time. The tests were conducted using *OpenSSL* on a virtual machine with Intel Core i7-6700 CPU (one core, 3.4 GHz) and 4 GB of memory.

Storage Overhead: Consider the DHCPv6 client on Ubuntu, for example. A DHCPv6 lease with one APA in the IA_TA requires about 480 B of storage, including interface, options, and APA information. If we assume that a host is assigned 100 APAs, the storage overhead is about 12 KB. For the APA Assigner, if we assume that an AS has 1 million hosts, it requires about 11.2 GB of disk space to store the APAs when APA Assigner assigns 100 APAs to each host. APNA uses certificates to certify the binding between an EphID and a public/private key pair. Currently, the most common certificate sizes on the Internet today vary between 500 - 1500 B [45]. Similarly, suppose we assume that there are 1 million hosts, and each sends 100 flows. In that case, APNA's management service will cost about 46 GB of disk space to store the EphID certificates, even with the size of each certificate being 500 B. APIP does not assign multiple addresses to hosts, but it has a privacy implication, i.e., the break of sender-flow unlinkability within an AS, when using network address translation.

2) APA Assignment Overhead: As we implemented the functionality of APA assignment based on IA_TA in DHCPv6, we evaluated the performance of the APA Assigner's response to a message from a host. When receiving a Solicit message, the APA Assigner used about 612 μs to build the Advertise message to respond to the host. When receiving a Request message, the APA Assigner used about 526 μs to build the Reply message for the host.



(a) Bit rate (b) Packet rate

Fig. 6. Verification at border routers.

3) Data Forwarding: As a border router checks the IPC of the PAVI header and masks the source and destination addresses, we evaluated its forwarding performance. We used two Dell commodity servers with two 2.40 GHz Intel Xeon E5-2620 v3 CPUs, 16 GB of memory, and four dual-port 10 Gb NICs (a total of 80 Gbps). One server worked as a border router and directly connected to the other packet generator server installed with MoonGen [46].

Bandwidth Overhead: We tested the forwarding performance of all ports for four packet sizes: 128, 256, 512, and 1024 bytes. Fig. 6 shows the results. The baseline for the experiments was the forwarding performance of legacy IPv6 traffic. The PAVI header processing throughput at the border routers matched well with the baseline because it only involved one MAC computation, two encryptions, and two lookups. If we assume that the average packet size was 540 bytes [47], sending packets with the PAVI header consumed an additional 6.25% of the bandwidth of the original traffic. However, when APNA is deployed on today's Internet, it inserts a Generic Routing Encapsulation header [48] (at least 4 bytes) and APNA header (56 bytes) into each packet, which consumes at least an additional 10% of the bandwidth.

Storage Overhead: The storage overhead has two aspects: shared keys with ASes, and shared keys with hosts. We assumed that the length of the keys shared with ASes was 128 bits (AES-128) and the length of key index was 4 bytes; then, the border router used about 1.25 MB to store keys shared with 65,536 ASes. For keys shared with hosts, if we assume that 1,000,000 users in the AS, the storage cost was about 19 MB.

4) Comparison With Other Protocols: As PAVI, APIP, and APNA all update border routers,[4] border routers' performance should not degrade. We evaluated the performance of border routers for the above three protocols. More specifically, we compared the throughput of a border router for APIP, APNA, and PAVI in today's Internet by sending different traffic rates to the border router. For APIP, based on the experimental setup of data forwarding described in Section VI-B.3, we used another Dell commodity server with two 2.40 GHz Intel Xeon E5-2620 v3 CPUs, 16 GB of memory to work as

[4]On the one hand, Design 1 of APIP in Section 7.4 will require the update of border routers. On the other hand, border routers of an AS can verify any flows that it forwards.

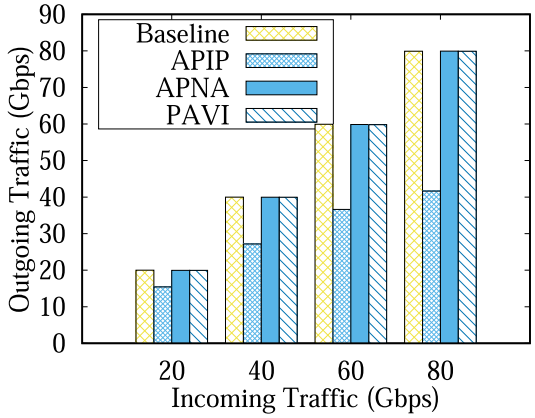


Fig. 7. Throughput comparison among APIP, APNA, and PAVI at border routers.

a delegate, which directly connected to the border router. The incoming traffic for APIP was set to contain ten million flows for five minutes. Simultaneously, the border router worked as a verifier and verified flows to the delegate before forwarding them. As APIP uses PKI to relax its assumption that HIDs are self-certifying in Section 7.3, we used OpenSSL to create self-signed PKI certificates for the delegate. The border router checked a delegate's certificate before trusting its response.[5] For APNA, we followed the implementation of the APNA border router described in Section 5.2.2 of APNA. We used the method provided in Section 7.2 of APNA [10] to simulate the communication process at border routers. We used the same experimental setup mentioned in Section VI-B.3 to evaluate an APNA border router's performance. The result is shown in Fig. 7. As the rate of incoming traffic increases, the border routers of APNA and PAVI both work well. However, the throughput of the APIP border router increases much slower than baseline. This is because the APIP border router needs to communicate with the delegate and verify its certificates. Also, we compared PAVI with APIP and APNA in terms of deployment cost. The result is shown in Table VI. Compared with APIP and APNA, PAVI does not require all the routers' updates but only the border routers' updates. As PAVI does not introduce new communication identifiers, modifications to infrastructures and protocols related to communication identifiers were not required, including routing protocols and DNS. However, APIP and APNA required these modifications.

VII. Discussion

A. Block Cipher Length

In PAVI, we can use either 64-bit or 128-bit block ciphers to generate APAs. The difference concerns whether to store extra information used to generate APAs. For example, when using the AES algorithm, an AS should store the IV for each APA to obtain UIDs from each APA thereafter. As each user subscribes to the privacy service, it is necessary for an AS to store IV values of each user. But if the AS chooses IDEA algorithm, it can use secret keys to decrypt APAs to obtain UIDs directly without storing IV values. However, the advantage of AES is that it is often accelerated by hardware [44].

[5]This is an inevitable process no matter which design in Section of APIP [9] the deployer AS adopts. In Design 1, the border router will check the certificates of other networks' border routers; in Design 2, the border router will check the third-party delegates' certificates.

B. Partial Deployment

Regardless of whether transit ASes deploy PAVI, they should forward packets with the PAVI header. If $\mathcal{A}_D$ does not deploy PAVI, it should at least support the decryption of the destination IID and EEA in the PAVI header. If $\mathcal{A}_D$ does not support destination IID decryption, source border routers do not mask the destination address of each packet. When an upgraded AS receives packets originating from a legacy AS, its border routers only forward these packets. In such a case, these packets can not be accountable.

C. Source and Destination Masking

The sender's masking destination addresses, the source border router's masking destination addresses, and the destination border router's masking source addresses can help increase sender-receiver unlinkability against observers in the source LAN segment, transit ASes, and observers in the destination LAN segment, respectively, especially when the receiver is a server that does not change its addresses frequently, the destination AS does not upgrade, or the sender uses the same APA to send multiple flows. If a server never changes its address, the sender-receiver unlinkability against the source AS will decrease to zero.

VIII. Related Work

Balancing Accountability and Privacy: Persona [49] balances accountability and anonymity for the next generation Internet (NGI). In Persona, routers encrypt the source IP address of each packet at each hop to hide the source identity of the packet. A victim can ask the last router to backtrace all routers by decrypting the IPs to provide accountability. Although Persona improves anonymity for NGI, it may not provide correct accountability due to route asymmetry and the lack of integrity protection for packets.

The Accountable and Private Internet Protocol (APIP) [9] balances accountability and privacy at the network layer. It splits source addresses into two fields: an accountability address and a return address. Accountability addresses point to *accountability delegates* that vouch for the source's packets. While accountability is handled by delegates, return addresses can be masked to protect the source's privacy. Sources send the fingerprints of packets to their accountability delegates, which work as proof of vouching when verifiers challenge them. However, when APIP is adapted to today's Internet (in Section 7.3 of APIP [9]), the verification of PKI certificates and the communication with delegates at verifiers will reduce verifiers' performance.

In the Accountable and Private Network Architecture (APNA) [10], ISPs link every packet to its sender without revealing the sender's identity through a cryptographic token called the ephemeral identifier (EphID). Privacy in APNA is not limited to host anonymity but extends to data confidentiality. However, it does not ensure data confidentiality. First, an AS can carry out a man-in-the-middle (MitM)

TABLE VI
DEPLOYMENT COMPARISON WITH OTHER PROTOCOLS

Protocol	Host	Access Router	Border Router	DNS	Intra-AS Routing	BGP
APIP [9]	✔	✔	✔	✔	✔	✔
APNA [10]	✔	✔	✔	✔	✔	✔
PAVI	✔	✘	✔	✘	✘	✘

* ✔ indicates upgrade required. ✘ indicates not required.

attack to decrypt communication between hosts in it, as the identities of both are already known to the AS. Second, APNA introduces receive-only EphIDs ($EphID_r$) to prevent the threat of attackers continuously sending shutoff requests against a victim domain. But nodes in the communication path between client and server can launch MitM attacks because the client never knows the $EphID_s$ used to establish a connection with the client by the server. Another issue is that large-scale modifications of DNS records, protocols (e.g., routing protocols), and infrastructures (e.g., all routers in ISPs) enhance the difficulty of deployment of APNA. Compared with APNA, PAVI does not require such modifications but the update of border routers. Because IPv6 is being deployed commonly around the world, many routers support IPv6 already. Even if not all border routers are upgraded, the observers can not undermine user privacy from APAs, unlike APNA. Compared with the previous version of this paper [1], we have made substantive enhancements in this manuscript. First, we give a more detailed design of the processing of outgoing and incoming packets at the border routers of the source and destination ASes, respectively. Second, we analyze privacy more thoroughly using different cases from different angles in Section V-B.2. We explore the influence of the number of users, APAs, and flows on sender-flow unlinkability, respectively. Finally, we compare PAVI with state of the arts in terms of storage, bandwidth, and forwarding overhead to demonstrate that PAVI is a lightweight and deployable protocol. PAVI neither reduces the performance of forwarding devices, nor does it require large-scale updates to the existing infrastructures and protocols.

Accountability: Some protocols, such as AaaS [50], AIP [51], BAFI [52], and IPA [53] have been proposed to address accountability. AaaS is the first scheme to consider accountability as a service but is costly. AIP and BAFI require host renumbering, and IPA provides prefix-level accountability. None of these protocols addresses privacy.

Privacy: Some mechanisms have been proposed to support privacy. The IPv6 Internet supports source anonymity to some extent through the use of temporary addresses [15] at the expense of accountability. Moreover, such anonymity is constrained to the day level because a host regenerates its temporary address every other day by default. IPsec [22] is supported naively in IPv6 and maintained by IETF, and can be used to ensure data confidentiality and integrity. Overlay systems such as onion routing [6], [7] and mix networks [8] are designed to provide anonymity at the cost of accountability.

IX. CONCLUSION

In this paper, we proposed PAVI, which bootstraps accountability and privacy in the IPv6 Internet. For the former, it allows user-level associations with packets. For the latter, it achieves sender anonymity, sender-flow unlinkability, sender-receiver unlinkability, and data confidentiality. The results of a quantitative analysis and performance evaluation show that PAVI can achieve accountability and privacy, and is a lightweight and deployable protocol.

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